MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 3C24

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode

See figure 1

Mounting position: Vertical, base down or up

Weight: 1.5-ounces nominal (42.5 grams)

ABSOLUTE RATINGS: C Telegraphy

Parameter:	Fl	Ef	Eb	Ec	Ib	Ic	Pp
Unit:	MHz	Vac	kVdc	Vdc	mAdc	mAdc	W
Maximum:	60	6.6	2.0	-500	75	25	25
Minimum:	---	6.0	---	---	---	---	---
TEST CONDITIONS:	---	6.3	1.0	Adjust	35	---	---

GENERAL:

Qualification - Required

(E) denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:05Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1231	Peak emission	eb = ec = 1,000 v	0.65	II	is	1.5	---	a
1261	Electrode voltage (grid)		0.65	II	Ec	-14	-24	Vdc
1266	Total grid current	See note	0.65	II	Ic	---	-5.0	uAdc
1301	Filament current		0.65	II	If	2.80	3.15	Aac
	Quality conformance inspection, part 2							
1037	Low-frequency vibration	Rp = 2,000 ohms; Eb = 250 Vdc; Ec/Ib = 10 mAdc	---	---	Eb	---	500	mVac
1036	Bump	Angle = 25°	---	---	---	---	---	---
1236	Power oscillation	F = 100 MHz; Eb = 1,000 Vdc; Ib = 80 mAdc; Ic = 25 mAdc	---	---	Po	45	---	W
1316	Amplification factor		---	---	Mu	22	28	---
1331	Direct-interelectrode capacitance	No shield	---	---	Cgp	1.4	1.8	pF
					Cin	1.4	2.2	pF
					Cout	0.1	0.3	pF
	Quality conformance inspection, part 3							
---	Life test	Group C; Pp = 25 W; t = 500 hours	---	---	---	---	---	---
---	Life-test end point:							
1231	Peak emission		---	---	is	1.0	---	a

NOTE: This test is to be the first test performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

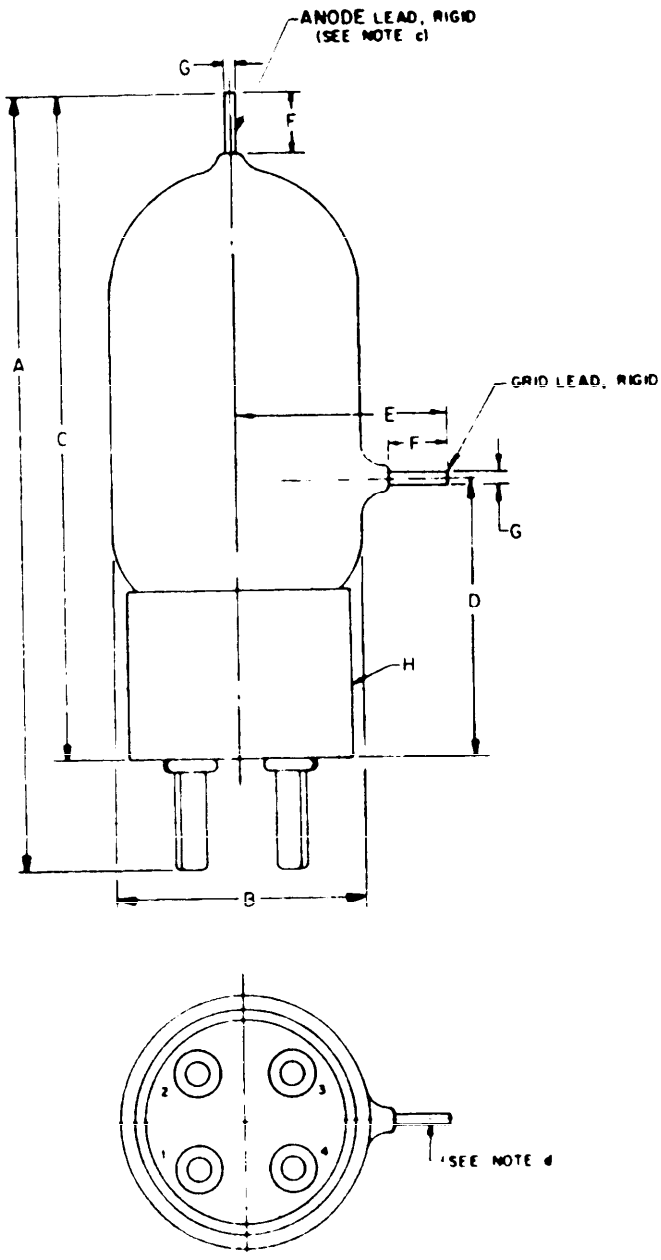
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2879)

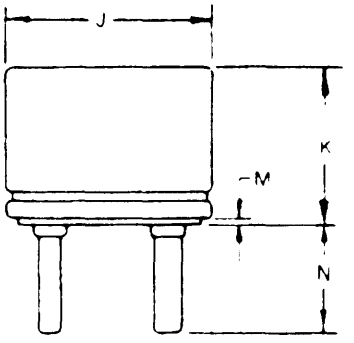
Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:05Z
Check the source to verify that this is the current version before use.



PIN CONNECTIONS	
Pin No.	Element
1	f
2	nc
3	nc
4	f
Anode lead	a
Grid lead	g

FIGURE 1. Outline drawing of electron tube type 3C24.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:05Z
Check the source to verify that this is the current version before use.



Alternate base, 4 pin, metal shell with ceramic wafer insert.

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	4.000 (101.60)	4.375 (111.13)
B	1.313 (33.35)	1.438 (36.53)
C	3.404 (86.46)	
D	1.437 (36.50)	1.687 (42.85)
E	1.100 (27.94)	1.300 (33.02)
F	.310 (7.87)	.440 (11.18)
G	.045 (1.14)	.051 (1.30)
Ⓔ Quality conformance inspection, part 3 (see note e)		
H	Base: A4-5 (see note b)	
J	1.137 (28.88)	1.165 (29.59)
K	.820 (20.83)	.850 (21.59)
Reference dimensions		
M	.032 (.81)	
N	.563 (14.30)	

NOTES:
a. For pin alinement, use gage GA4-1.
b. A metal shell base with ceramic wafer insert may be used.
c. Deviation of centerline of anode lead from centerline of base shall not exceed .125 (3.18 mm).
d. Deviation of centerline of grid lead from centerline of base shall not exceed 3 .
e. These dimensions shall be tested annually, with the following sampling plan:

n1 = 4 c1 = 0 where c2 represents the total allowable
n2 = 4 c2 = 1: failures for the first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become quality conformance inspection (QCI), part 2, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3 basis.

FIGURE 1. Outline drawing of electron tube type 3C24 - Continued.

☆ U. S. GOVERNMENT PRINTING OFFICE: 1975 603-114 3727

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:05Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:05Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

[ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:05Z
Check the source to verify that this is the current version before use.